

## Appendix

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The appendix is organized into three sections:

- Section **A** provides the pseudocode of Talk-GRPO, including rollout sampling and GRPO-based policy optimization.
- Section **B** describes the experimental setup, including dataset splits, baseline methods, checkpoints, and implementation details.
- Section **C** provides additional analyses, including training dynamics and the stage-wise effects of GRPO post-training and temporal refinement.

## A Pseudocode

### A.1 Talk-GRPO Training Procedure

We divide **Talk-GRPO** into two phases: rollout sampling and policy optimization. The two phases are summarized in algorithm 1 and algorithm 2.

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#### Algorithm 1 Talk-GRPO Sampling Phase

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**Require:** Pretrained generator  $\pi_\theta$ , dataset  $\mathcal{D}$ , group size  $G$ , sampling steps  $T$ .

**Ensure:** Rollout buffer  $\mathcal{B}$ .

- 1: Initialize rollout buffer  $\mathcal{B}$ .
- 2: Sample an input-target pair  $(\mathbf{c}, x^*) \sim \mathcal{D}$ , where  $\mathbf{c} = (x^{\text{ref}}, L^d, \mathbf{a})$ .
- 3: Set  $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta$ .
- 4: **for**  $i = 1$  **to**  $G$  **do**
- 5:     Sample a generation trajectory

$$\tau_i = \{x_T^i, \dots, x_0^i\} \sim \pi_{\theta_{\text{old}}}(\cdot \mid \mathbf{c}).$$

- 6:     **for**  $t = T$  **to**  $1$  **do**
- 7:         Store the old-policy log-probability for the current transition:

$$\ell_{i,t}^{\text{old}} = \log p_{\theta_{\text{old}}}(x_{t-1}^i \mid x_t^i, \mathbf{c}).$$

- 8:     **end for**
- 9:     Obtain the final generated output and Extract facial landmarks

$$\hat{x}_i = x_0^i, \quad \hat{L}_i = \mathcal{M}(\hat{x}_i).$$

- 10:     Compute the terminal reward

$$R_i = R(\hat{x}_i, L^d, x^*).$$

- 11: **end for**
- 12: Compute group reward statistics

$$\mu_R = \frac{1}{G} \sum_{i=1}^G R_i, \quad \sigma_R = \sqrt{\frac{1}{G} \sum_{i=1}^G (R_i - \mu_R)^2}.$$

- 13: **for**  $i = 1$  **to**  $G$  **do**
- 14:     Compute the group-relative advantage

$$A_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon_{\text{adv}}}.$$

- 15:     Add

$$(\mathbf{c}, \tau_i, \{\ell_{i,t}^{\text{old}}\}_{t=1}^T, A_i)$$

to  $\mathcal{B}$ .

- 16: **end for**
  - 17: **return**  $\mathcal{B}$ .
-

**Algorithm 2** Talk-GRPO Training Phase

**Require:** Rollout buffer  $\mathcal{B}$ , policy  $\pi_\theta$ , frozen reference policy  $\pi_{\text{ref}}$ , inner epochs  $J$ , clipping threshold  $\epsilon_{\text{clip}}$ , KL weight  $\lambda_{\text{KL}}$ .

**Ensure:** Updated generator  $\pi_\theta$ .

1: **for**  $j = 1$  **to**  $J$  **do**

2:   Sample

$$\left(\mathbf{c}, \tau_i, \{\ell_{i,t}^{\text{old}}\}_{t=1}^T, A_i\right)$$

from  $\mathcal{B}$ .

3:   Sample a denoising step  $t$  from trajectory  $\tau_i$ .

4:   Compute the current transition log-probability:

$$\ell_{i,t}(\theta) = \log p_\theta \left( x_{t-1}^i \mid x_t^i, \mathbf{c} \right).$$

5:   Compute the policy ratio:

$$r_{i,t}(\theta) = \exp \left( \ell_{i,t}(\theta) - \ell_{i,t}^{\text{old}} \right).$$

6:   Compute the clipped GRPO loss:

$$\begin{aligned} \mathcal{L}_{\text{GRPO}} = & -\min \left[ r_{i,t}(\theta) A_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) A_i \right] \\ & + \lambda_{\text{KL}} D_{\text{KL}} \left( p_\theta(\cdot \mid x_t^i, \mathbf{c}) \parallel p_{\text{ref}}(\cdot \mid x_t^i, \mathbf{c}) \right). \end{aligned}$$

7:   Update  $\theta$  by minimizing  $\mathcal{L}_{\text{GRPO}}$ .

8: **end for**

9: **return**  $\pi_\theta$ .

**B Experimental Setups****B.1 Dataset Details**

Table 5: Dataset statistics used in our experiments. MEAD is used for training and testing, while HDTF and CREMA-D are used only as independent cross-dataset test sets.

| Split | MEAD | HDTF | CREMA-D | Duration   |
|-------|------|------|---------|------------|
| Train | 720  | —    | —       | 1 hour     |
| Test  | 360  | 360  | 150     | 1.21 hours |

We provide additional details about the datasets and preprocessing used in our experiments. All videos are split into fixed-length 5-second clips and sampled at 25 FPS, resulting in 125 frames per clip. This setting provides a consistent temporal length and facilitates frame-level alignment between speech features and facial landmarks. We use MEAD for post-training and evaluate on MEAD, HDTF, and CREMA-D. To explicitly study the effectiveness of Talk-GRPO in a low-data post-training regime, we deliberately use only approximately one hour of MEAD data for training. For MEAD, we adopt an identity-disjoint train/test split to evaluate generalization to unseen identities, while HDTF and CREMA-D are used only as independent test sets for cross-dataset evaluation. The detailed data split is summarized in Table 5.

## B.2 Baseline Details

Table 6: Summary of compared talking-face generation methods.

| Method           | Supervision | Region     | Generator | Size |
|------------------|-------------|------------|-----------|------|
| Hallo2 [7]       | MSE         | Full face  | Diffusion | 512  |
| MEMO [30]        | MSE         | Full face  | Diffusion | 512  |
| EchoMimic [4]    | MSE + LPIPS | Full face  | Diffusion | 512  |
| Diff2Lip [13]    | SyncNet     | Lower half | Diffusion | 128  |
| TalkLip [19]     | AV-HuBERT   | Lower half | GAN       | 96   |
| <b>Talk-GRPO</b> | Lmk + LPIPS | Full face  | Diffusion | 512  |

Table 6 summarizes the compared methods. We group the comparison methods into two categories. The first includes full-face diffusion-based methods, such as Hallo2, MEMO, and EchoMimic, which primarily rely on diffusion-based training objectives and generate full-face videos. The second includes methods with learned synchronization supervision, such as Diff2Lip and TalkLip, which employ SyncNet, or AV-HuBERT and typically operate on cropped mouth or lower-face regions. **Talk-GRPO** builds on a full-face diffusion generator and performs post-training using explicit landmark-based rewards. Unlike learned synchronization-supervised methods, it does not require an additional learned synchronization model or cropped lower-face supervision while directly generating full-resolution facial videos.

**Checkpoints.** For all baseline methods, we use the publicly released pretrained checkpoints provided by the original authors whenever available. **Talk-GRPO** is initialized from the pretrained EchoMimic checkpoint and further post-trained with our two-stage framework. Unless otherwise stated, we use the checkpoint post-trained with full-landmark conditioning for evaluation. During inference in the **Landmark w/o Mouth** setting, the mouth landmarks are removed and audio is provided to drive mouth motion, while the remaining landmarks control head pose and non-mouth facial motion.

## B.3 Implementation Details

Table 7 summarizes the main implementation settings of **Talk-GRPO**. We use EchoMimic [4] as the pretrained base generator and conduct Stage 1 post-training on two NVIDIA A100 GPUs. Inference is performed on a single NVIDIA RTX 4090 GPU. EchoMimic is used as the base generator. Stage 1 performs GRPO post-training with group size  $G = 4$ , while Stage 2 applies supervised temporal refinement with  $N = 4$  temporal samples. For a fair comparison, the supervised fine-tuning baselines are trained with the same number of optimization iterations as Talk-GRPO, so that the observed differences are not simply due to a larger training budget. To calculate landmark-based reward, we use MediaPipe to extract 468 facial landmarks from each generated frame and select 40 mouth-region landmarks (2D), highlighted by the white contour, to compute the mouth landmark distance used in the landmark-based reward.

Table 7: Implementation details of Talk-GRPO.

| Setting                                        | Value                          |
|------------------------------------------------|--------------------------------|
| Base generator                                 | EchoMimic                      |
| Post-training hardware                         | $2 \times$ NVIDIA A100 GPUs    |
| Inference hardware                             | $1 \times$ NVIDIA RTX 4090 GPU |
| Sampling steps                                 | 40                             |
| <i>Stage 1: GRPO Post-Training</i>             |                                |
| Group size $G$                                 | 4                              |
| GRPO inner epochs $J$                          | 2                              |
| KL weight $\lambda_{KL}$                       | 0.04                           |
| MediaPipe landmarks                            | 468 points                     |
| Mouth landmarks                                | 40 points                      |
| Reward weight $w_{\text{mouth}}$               | 0.8                            |
| Reward weight $w_{\text{LPIPS}}$               | 0.2                            |
| Condition dropout ratio $r$                    | 0.5                            |
| Training time                                  | $\sim 2$ days                  |
| <i>Stage 2: Temporal Refinement</i>            |                                |
| Temporal samples $N$                           | 4                              |
| Perceptual loss weight $\lambda_{\text{perc}}$ | 0.15                           |
| Training time                                  | $\sim 2$ days                  |
| Supervised baseline training                   | Matched iterations             |

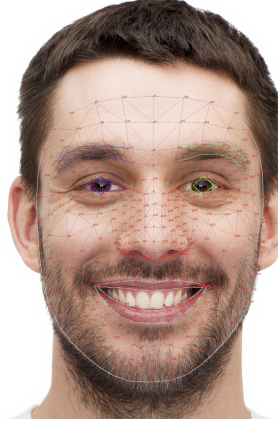


Fig. 3: MediaPipe facial landmark visualization.

## C Extended Results and Analysis

### C.1 Training Dynamics and Loss Curves

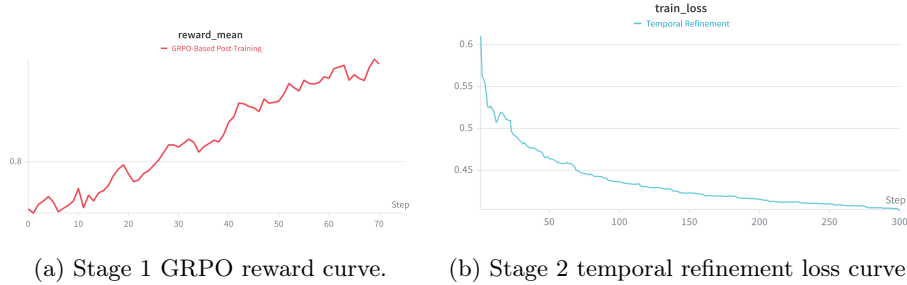


Fig. 4: Training dynamics of the two-stage **Talk-GRPO** framework. Stage 1 shows the mean reward during GRPO-based post-training, while Stage 2 shows the supervised temporal refinement loss. The increasing reward in Stage 1 and the decreasing loss in Stage 2 indicate stable optimization of the two complementary objectives.

Figure 4 shows the optimization behavior of the two stages in Talk-GRPO. In Stage 1, the mean reward gradually increases over 75 steps, despite small fluctuations caused by group sampling and reward normalization. This indicates that the generator progressively learns to produce candidates with higher landmark-based rewards, corresponding to better mouth-region alignment and

image-preservation quality. The reward curve does not collapse or oscillate severely, suggesting that the clipped GRPO objective and KL regularization provide stable post-training behavior.

In Stage 2, the temporal refinement loss decreases steadily over 300 optimization steps. The loss drops rapidly in the early stage and then continues to decrease more smoothly, indicating stable convergence of the supervised temporal refinement objective. Together, these curves show that Stage 1 and Stage 2 optimize complementary goals: GRPO post-training improves the final-output landmark-based reward, while temporal refinement further improves video-level consistency.

## C.2 Stage-wise Analysis of GRPO and Supervised Finetuning

Table 8 shows that the two stages contribute differently to the final performance. Stage 1 GRPO post-training mainly improves landmark fidelity, reducing MLD by approximately 15.6% on MEAD and 20.2% on HDTF, with a smaller improvement of about 6.1% on CREMA-D. In contrast, its effect on temporal stability is limited. After Stage 2, FLVD decreases substantially by more than 40% on all three datasets, while MLD is further improved. Compared with the supervised fine-tuning baselines, Talk-GRPO achieves the lowest MLD and FLVD after both stages on all three datasets. These results indicate that Stage 1 primarily improves landmark fidelity through reward-based post-training, whereas Stage 2 is mainly responsible for temporal refinement.

Table 8: Detailed stage-wise ablation on MEAD, HDTF, and CREMA-D. *Initial* reports MLD and FLVD before adaptation. In Stage 1, Talk-GRPO uses landmark-based GRPO post-training, while AniPortrait and EchoMimic use supervised fine-tuning with matched training iterations. Stage 2 further applies temporal refinement. Best results within each stage are shown in **bold**.

| Dataset | Model            | Objective | Initial |        | Stage 1       |               |               |               | Stage 2       |               |               |              |
|---------|------------------|-----------|---------|--------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|
|         |                  |           | MLD↓    | FLVD↓  | MLD↓          | FID↓          | SSIM↑         | FLVD↓         | MLD↓          | FID↓          | SSIM↑         | FLVD↓        |
| MEAD    | AniPortrait      | MSE       | 16.883  | 8.330  | 15.672        | 23.495        | <b>91.297</b> | <b>8.019</b>  | 14.030        | <b>21.189</b> | 90.830        | 6.184        |
|         | EchoMimic        | MSE+LPIPS | 17.342  | 9.425  | 16.073        | <b>21.363</b> | 89.989        | 9.310         | 15.099        | 26.325        | 89.492        | 5.479        |
|         | <b>Talk-GRPO</b> | Lmk+LPIPS | 17.342  | 9.425  | <b>14.628</b> | 21.605        | 90.017        | 9.224         | <b>13.312</b> | 24.036        | <b>91.343</b> | <b>5.147</b> |
| HDTF    | AniPortrait      | MSE       | 16.531  | 10.913 | 15.271        | 29.743        | 84.843        | <b>10.774</b> | 15.267        | 29.337        | 83.120        | 8.189        |
|         | EchoMimic        | MSE+LPIPS | 17.763  | 12.483 | 16.252        | <b>23.079</b> | 86.750        | 12.124        | 14.938        | 24.701        | 86.749        | 7.319        |
|         | <b>Talk-GRPO</b> | Lmk+LPIPS | 17.763  | 12.483 | <b>14.178</b> | 23.561        | <b>86.839</b> | 12.036        | <b>14.009</b> | <b>23.451</b> | <b>86.836</b> | <b>7.167</b> |
| CREMA-D | AniPortrait      | MSE       | 15.298  | 8.344  | <b>13.655</b> | 34.437        | <b>87.980</b> | <b>8.250</b>  | 13.184        | 35.519        | <b>87.181</b> | 6.098        |
|         | EchoMimic        | MSE+LPIPS | 16.664  | 10.519 | 15.756        | <b>27.791</b> | 85.904        | 10.430        | 13.676        | 31.119        | 85.937        | 5.506        |
|         | <b>Talk-GRPO</b> | Lmk+LPIPS | 16.664  | 10.519 | 15.644        | 29.536        | 85.972        | 10.355        | <b>13.017</b> | <b>29.788</b> | 86.934        | <b>5.340</b> |